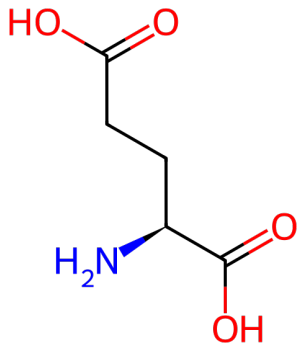
MOLECULE

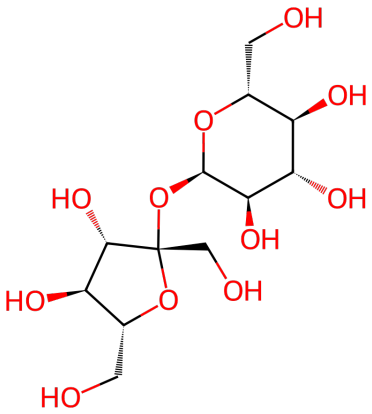


(2S)-2-Aminopentanedioic acid

$C_5H_9NO_4$

MolNeuro Retreat 2026

MOLECULE

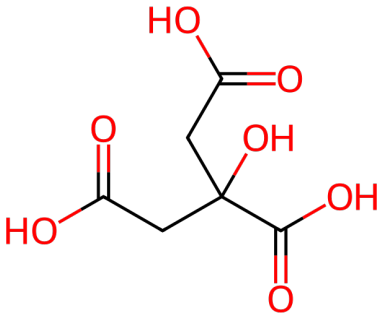


β-D-Fructofuranosyl α-D-glucopyranoside

$C_{12}H_{22}O_{11}$

MolNeuro Retreat 2026

MOLECULE



2-Hydroxypropane-1,2,3-tricarboxylic acid

$C_6H_8O_7$

MolNeuro Retreat 2026

MOLECULE

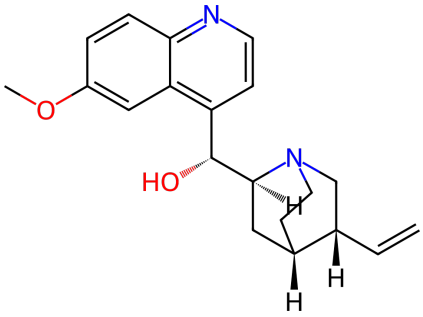
Na^+ Cl^-

Sodium chloride

NaCl

MolNeuro Retreat 2026

MOLECULE

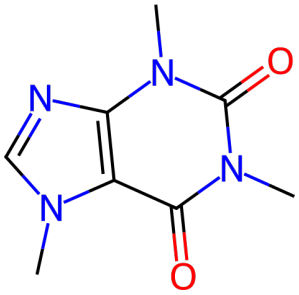


(8S,9R)-6'-Methoxycinchonan-9-ol

$C_{20}H_{24}N_2O_2$

MolNeuro Retreat 2026

MOLECULE

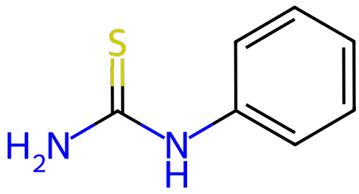


1,3,7-Trimethylxanthine

$C_8H_{10}N_4O_2$

MolNeuro Retreat 2026

MOLECULE

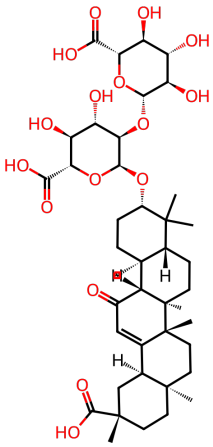


Phenylthiocarbamide (PTC)

$C_7H_8N_2S$

MolNeuro Retreat 2026

MOLECULE

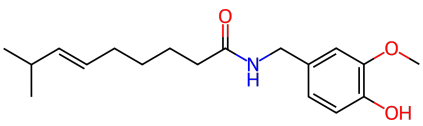


Glycyrrhizin

$C_{42}H_{62}O_{16}$

MolNeuro Retreat 2026

MOLECULE

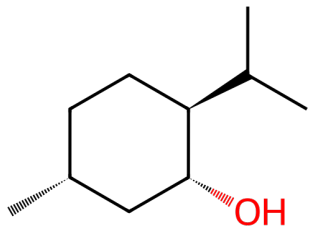


(6E)-N-(4-Hydroxy-3-methoxybenzyl)-8-methylnon-6-enamide

$C_{18}H_{27}NO_3$

MolNeuro Retreat 2026

MOLECULE

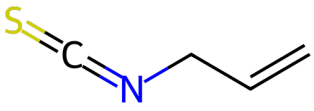


(1R,2S,5R)-2-Isopropyl-5-methylcyclohexan-1-ol

$C_{10}H_{20}O$

MolNeuro Retreat 2026

MOLECULE

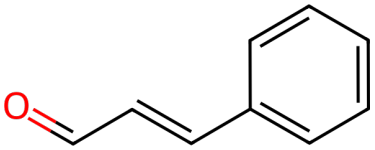


3-Isothiocyanatoprop-1-ene

C_4H_5NS

MolNeuro Retreat 2026

MOLECULE

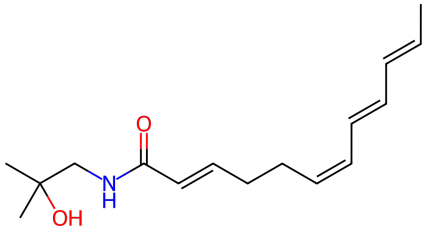


(2E)-3-Phenylprop-2-enal

C_9H_8O

MolNeuro Retreat 2026

MOLECULE

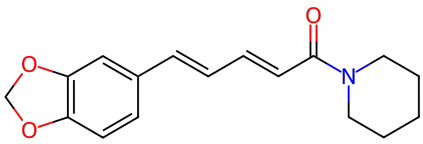


(2E,6Z,8E,10E)-N-(2-Hydroxy-2-methylpropyl)dodeca-2,6,8,10-tetraenamide

$C_{16}H_{25}NO_2$

MolNeuro Retreat 2026

MOLECULE

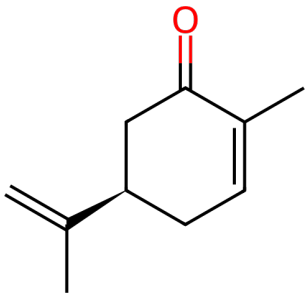


(2E,4E)-5-(1,3-Benzodioxol-5-yl)-1-(piperidin-1-yl)penta-2,4-dien-1-one

$C_{17}H_{19}NO_3$

MolNeuro Retreat 2026

MOLECULE

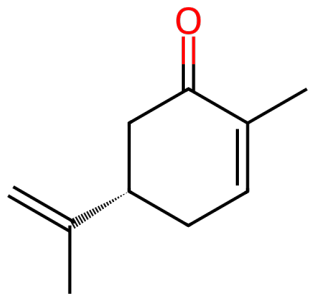


(R)-(−)-Carvone

$C_{10}H_{14}O$

MolNeuro Retreat 2026

MOLECULE

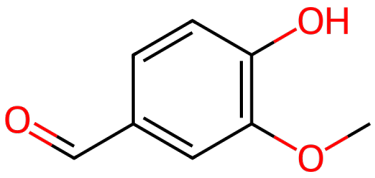


(S)-(+)-Carvone

$C_{10}H_{14}O$

MolNeuro Retreat 2026

MOLECULE

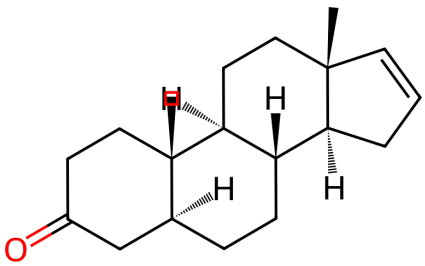


4-Hydroxy-3-methoxybenzaldehyde

$C_8H_8O_3$

MolNeuro Retreat 2026

MOLECULE

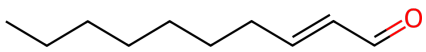


Androstenone

$C_{19}H_{28}O$

MolNeuro Retreat 2026

MOLECULE

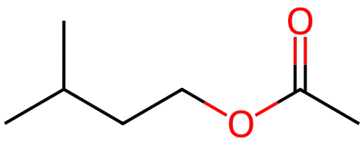


(E)-2-Decenal

$C_{10}H_{18}O$

MolNeuro Retreat 2026

MOLECULE



3-Methylbutyl acetate

$C_7H_{14}O_2$

MolNeuro Retreat 2026

FLAVOUR



Umami

Parmesan · tomato · kombu

RECEPTOR

TAS1R1 / TAS1R3

MolNeuro Retreat 2026

FLAVOUR



Sweet

Sugar cane · sugar beet · sweets

RECEPTOR

TAS1R2 / TAS1R3

MolNeuro Retreat 2026

FLAVOUR



Sour

Lemon · lime

RECEPTOR

OTOP1 (proton channel)

MolNeuro Retreat 2026

FLAVOUR



Salty

Sea salt · crisps

RECEPTOR

ENaC

MolNeuro Retreat 2026

FLAVOUR



Bitter

Tonic water

RECEPTOR

Several TAS2Rs

MolNeuro Retreat 2026

FLAVOUR



Bitter

Coffee · tea

RECEPTOR

TAS2R7/10/14/43/46

MolNeuro Retreat 2026

FLAVOUR



Bitter... or nothing?

Broccoli · Brussels sprouts

RECEPTOR

TAS2R38

MolNeuro Retreat 2026

FLAVOUR



Intensely sweet

Liquorice root · salmiak

RECEPTOR

TAS1R2 / TAS1R3

MolNeuro Retreat 2026

FLAVOUR



Burning heat

Chilli peppers

RECEPTOR

TRPV1

MolNeuro Retreat 2026

FLAVOUR



Cooling

Peppermint · cough drops

RECEPTOR

TRPM8

MolNeuro Retreat 2026

FLAVOUR



Sharp, up-the-nose

Wasabi · mustard · horseradish

RECEPTOR

TRPA1

MolNeuro Retreat 2026

FLAVOUR



Warm & spicy

Cinnamon · kanelbullar

RECEPTOR

TRPA1 (+ olfactory receptors)

MolNeuro Retreat 2026

FLAVOUR



Tingling & numbing

Sichuan pepper

RECEPTOR

KCNK3/9/18 (two-pore K^+ channels)

MolNeuro Retreat 2026

FLAVOUR



Peppery bite

Black pepper · cacio e pepe

RECEPTOR

TRPV1 / TRPA1

MolNeuro Retreat 2026

FLAVOUR



Spearmint

Spearmint · chewing gum

RECEPTOR

Olfactory receptors

MolNeuro Retreat 2026

FLAVOUR



Caraway & dill

Dill · caraway · aquavit

RECEPTOR

Olfactory receptors

MolNeuro Retreat 2026

FLAVOUR



Vanilla

Vanilla pods · ice cream

RECEPTOR

OR10G4

MolNeuro Retreat 2026

FLAVOUR



Urine? Sandalwood? Nothing?

Boar meat ('boar taint')

RECEPTOR

OR7D4

MolNeuro Retreat 2026

FLAVOUR



Fresh herb… or soap?

Coriander leaves

RECEPTOR

OR6A2 locus

MolNeuro Retreat 2026

FLAVOUR



Banana / pear drops

Bananas · pear-drop sweets

RECEPTOR

Olfactory receptors

MolNeuro Retreat 2026